

Review of MS-25325

This paper evaluates the effects of genomic testing (GT) technologies introduced in 2008 in the dairy industry. As shown in previous research cited in this paper, this technology has enabled significantly faster genetic improvements in dairy cattle breeding, by enabling much faster identification of dairy bulls with desirable genetics which lead to productivity improvements in their female offspring (dairy cows). However, the paper's primary empirical contribution is to highlight a major downside of GT technology—increased inbreeding in dairy cattle. To estimate these effects, the authors use a differences-in-differences (DID) econometric model combined with propensity score matching (PSM), to compare dairy bull inbreeding rates in highly popular dairy bull lineages v. matched control lineages before and after the widespread use of GT by 2010. After finding that GT significantly increased inbreeding within popular lineages, the paper attempts a benefit-cost analysis (BCA) of GT, by assessing the economic costs of the estimated inbreeding effects relative to the benefits of productivity improvements facilitated by GT.

Finding that GT increased inbreeding rates in dairy cattle strikes me as a significant contribution, not only in dairy science and economics but also in broader debates about unanticipated consequences of novel agricultural biotech. Indeed, I was left wondering if there might be similar concerns for genomics-assisted crop breeding, which is being increasingly advocated and deployed by plant scientists (e.g. Varshney et al. 2021): For example, might there be similar risks in terms of reduced genetic diversity from more precise and rapid trait selection in crop variety improvements? Might we wish that we had retained seemingly useless or deleterious genes in the population, to withstand unforeseen environmental conditions and stressors in the future? (I do not think the authors need to answer these big questions—but this paper could provide a specific, empirical example of why we might want to think more seriously about these questions.)

That said, there are several areas of the paper where major improvements are needed. Overall, a cross-cutting critique is that it feels like the authors assume too much prior knowledge of dairy industry and breeding than I think can be fairly expected of a broad AJAE readership—which I think could severely limit its appeal to this audience. The authors make an effort at increasing the breadth of their contribution by drawing a comparison with pesticide resistance, but as I explain below this comparison is superficial, and I think there are more direct, substantive ways at achieving this breadth.

Major concerns

- **Inbreeding as an externality?** The paper argues that the costs of inbreeding are an externality borne by the dairy industry as a whole, but that individual dairy bull

breeders do not have an incentive to consider these costs in their individual breeding decisions. The second section of the paper attempts to provide a descriptive conceptual framework for this argument, repeatedly comparing inbreeding to pesticide resistance management (specifically, Bt crops and refugia). However, I find the comparison to Bt resistance tenuous and more broadly am not yet convinced that the costs of inbreeding are an externality at all. Rather than using a descriptive analogy to argue their point, I think AJAE readers would be more convinced by a clear, concise theoretical model: Conceptually, pesticide resistance has been well-established through theoretical modeling (e.g. Hueth and Regev 1974; Miranowski and Carlson 1986; Cho et al. 2024) to present a type of common-pool resource (CPR) problem for farmers, whereby any farmer may use pesticides and draw upon the genetic pool of susceptibility in the pest population (non-excludability), but in doing so decreases the pool available to all farmers using it in the future due to resistance selection (rivalry). These features are borne out and clearly illustrated in bioeconomic models of pesticide resistance. With dairy cattle inbreeding, the nature and existence of the externality is less clear: Where is nonexcludability (or nonrivalry) present? Based on the information in the paper, high discount rates could provide a simple alternative explanation for increased inbreeding: Breeders may simply favor the short-run trait gains over the long-run risks of inbreeding, even if these discounted future risks are internalized in the market. Of course, dairy industry discount rates may likely be higher than is socially optimal (Wuepper et al. 2023), but this does not mean there is necessarily an externality present. One might also construe inbreeding as a type of pecuniary externality rather than a ‘real’ externality, in which competition and market forces among dairy bull breeders have negative consequences—albeit internalized and market-transmitted—on dairy producers (limited variety of non-inbred bull semen, e.g. in the event of a negative homozygous trait coming to light).

Part of the difficulty I have seeing the externality argument lies in the vague description of the risks of inbreeding and their costs. The main long-run risk the authors describe in section 2 is an increased “likelihood of unforeseen genetic traits that could harm the industry”... such as? That is all readers get, and they might be left wondering (as I was) why there’s not an incentive for improvements in GT technologies to better identify and predict these negative traits? A quick literature search (I have no background in dairy science or economics) revealed that such efforts may in fact be underway (Howard et al. 2017). This vagueness about the costs of inbreeding persists in Section 6 (Cost-Benefit Analysis), which uses an old estimate of the marginal costs of inbreeding (24 USD per 1% increase in inbreeding),

without any explanation of the basis for that figure and how it was calculated (does it depend on a discount rate?). Since it plays a pivotal role in the paper's economic conclusions, this figure deserves more attention and validation (ideally, this paper would generate an updated estimate of the marginal cost of inbreeding, though I understand that may be beyond the paper's scope).

- **Empirical methods and identification.** Overall, the evidence provided seems pretty clear that GT dramatically increased inbreeding rates. However, I do have some concerns, questions, and suggestions regarding the econometric methods used. The paper adopts a frequently used DID-PSM approach, which is compelling, but their setup makes a subtle argument in the definition of the treatment group, with implications for exactly what is being causally estimated: The authors define the treatment group of bulls as those coming from lineages of “superstar” founder bulls from 1998-2004, those whose descendants represented a very high proportion of all future bulls. They compare these superstar founders to those from matched controls who were not superstars but had comparable (measured) genetic traits. They then examine (using DID) how inbreeding rates changed between these two groups after the advent of GT technology. Notably, the treatment group is not defined as those bull lineages in which GT was adopted (the authors do not observe this information, and argue that GT was so widespread as to make estimation of GT treatment effects infeasible). So, what *do* these causal estimates tell us? Assuming the PSM is successful, this makes selection of trait-matched lineages into superstar status as good as random. Within the subsample of superstar and matched lineages, the treatment here must technically be defined as: GT (post-2010) x (randomly selection into) superstar status. But to the extent that it also increased inbreeding among non-superstars, this estimation can't reveal the aggregate effect of GT on inbreeding. Indeed, Figure 7 appears to show that inbreeding in the matched controls increased more rapidly post-GT (especially starting around 2014).

These subtle differences matter for the paper's conclusions and economic implications. Yet the BCA in section 6 seems to sweep these subtleties aside- the superstar-specific GT effect is used to calculate the aggregate, cumulative effect of GT on inbreeding, which is not correct given the definition of the treatment in the econometric model. In particular, it omits the costs of increased inbreeding in non-superstars.

Relatively minor points:

- Trait improvement equation (1): I don't think equation (1) serves any substantive purpose in the paper. Economists readers won't be able to make much of its utility (I wasn't at least, and I have some knowledge of genetics, though none of dairy bull breeding)—the units in this equation are unclear to me, and it's not linked to any theoretical model that might shed light on the incentives in the industry in relation to the alleged externality. Nor is it used in any of the econometric estimation (as far as I can tell).
- Examining Figures 7 and 8, I wonder if we are observing that GT led to earlier increases in inbreeding in superstars v. matched controls (perhaps due to earlier GT adoption?) with a possible leveling-out starting in 2019 (last year observed in data). One could speculate that after 2019, the non-superstars' inbreeding rates eventually caught up with those of superstars, perhaps as GT use further penetrated down into more marginal lineages. Indeed, one might expect that breeders of the control bulls—having similar genetic traits but less popular for purely random reasons—might eventually realize they could use GT to more rapidly improve their lines, communicate that value to dairy producers, and gain more market share.
- Line-level clustered standard errors. First, please report the number of lines (clusters) in the estimation table (table notes are also needed throughout- for *'s, standard errors, reference to corresponding model equations, etc.). Also, one could argue that firm-clustering may be desirable as well: You could do both (as long as lines weren't fully contained within firms, or vice versa), using multiway clustering!
- BCA in Section 6- discounting? It isn't clear how the inbreeding USD costs and genetic trait benefits are aggregated over the time period considered, and if a discount rate was used in that calculation. Moreover, if the costs of inbreeding include longer term risks (see above comment), it isn't clear how the discount rate might affect these calculations. Your BCA warrants its own appendix section detailing these calculations and assumptions for readers.
- PSM balance assessment & alternative matching estimators. From my understanding of the matching literature, raw t-stats and p-values in your balance table are not the best way to assess balance; 'mean standardized differences' (MSD) seem to have gained prominence as a way of assessing balance, and I personally much prefer a plot of MSDs, rather than tabular format (see Greifer 2025). Why did you only consider PSM in your matching strategy rather than alternatives such as nearest-neighbor, or inverse-probability weighting on the propensity score? If they haven't already done so, I encourage the authors to read

through the matching chapter of the Causal Inference Mixtape textbook (Cunningham 2021). Also, considering that superstar status in founder bulls might be related to firm-level marketing effort, can you include a firm-fixed effect in the propensity-score mode? Even better, in my opinion, is a within-cluster(firm) matching estimator, where matches are drawn from the same firm, if available, before drawing matches from other firms (Arpino and Cannas 2016).

- In the main text's description of the estimation models, some of the notation was hard to follow with incomplete definitions (e.g. *inb*, *yob*). I had to hunt around and deduce indirectly from the text to deduce that k indexes firm and α_k was the firm-level fixed effect (there's also a typo in the appendix, with α_j used in one instance where α_k should be).
- Figure 5: Because of the overlapping shading, I found this figure hard to read. One solution could be to use different line dashing / markers for the different overlaid density plots.
- In general, I think there are too many citations to the pesticide resistance, areawide pest control, and invasive species literatures (related to my comment above that I think the comparison here is misplaced).

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